

LOU BAKER, PhD

SUMMARY

Educator and researcher with expertise in image processing, data analysis, and experimental fluid mechanics. Dedicated to teaching, mentorship, and improving equity and accessibility in academia.

EDUCATION

University of Minnesota, Minneapolis, MN

Ph.D., Aerospace Engineering and Mechanics | 4.00

Jun 2021

Focus: Experimental Fluid Mechanics

M.S., Aerospace Engineering and Mechanics | 4.00

Dec 2018

B.Eng., Aerospace Engineering and Mechanics | 3.75

May 2016

EXPERIENCE

Financial Secretary & Field Organizer, UAW Local 4121, Seattle, WA Aug 2023 – present
Organizing for equity and accessibility for academic workers at the University of Washington.
Managing the Local's staff, finances, and database.

Teaching

Part-time Lecturer, Northeastern University, Seattle, WA May – Jun 2025
ME 3465: Introduction to Flight

Guest Lecturer, University of Washington, Seattle, WA May 2022
ME 543: Fluid Turbulence

Laboratory Instructor, University of Minnesota, Minneapolis, MN Sep 2016 – May 2017
AEM 4601: Instrumentation Laboratory
AEM 4602: Aeromechanics Laboratory

Teaching Assistant, University of Minnesota, Minneapolis, MN Jan – May 2014
AEM 2012: Dynamics

Research

NSF Postdoctoral Fellow, University of Washington, Seattle, WA Jun 2021 – Jul 2023
Environmental Fluid Mechanics Group, Mechanical Engineering Dept
Investigated microplastic particle dynamics in wavy, wind-driven surface flows

Graduate Research Assistant, University of Minnesota, Minneapolis, MN May 2016 – Jun 2021
St. Anthony Falls Laboratory, Aerospace Engineering & Mechanics Dept
Explored fluid-particle dynamics in turbulent open channel flows

Undergraduate Research Assistant, University of Minnesota Jan 2014 – May 2016
Aerospace Engineering & Mechanics Dept
Studied particle clustering in homogeneous, isotropic turbulence

LOU BAKER, PhD

Intern , Virgin Orbit, CA	Jun – Aug 2015
Designed engine gimbals and heat exchangers for LauncherOne rocket engines	
Intern , NASA Jet Propulsion Laboratory, CA	Jun – Aug 2013 & 2014
Designed and tested magnetic shielding for the ISS Cold Atom Lab research facility	

GRANTS & AWARDS

NSF Ocean Sciences Postdoctoral Research Fellowship	Aug 2021 – Aug 2023
National Defense Science and Engineering Graduate Fellowship	Sep 2017 – Jun 2021
Edward Silberman Fellowship	Jan 2019 – May 2020
John A. & Jane Dunning Copper Fellowship	May 2017
Albert George Oswald Prize for Outstanding Research	Sep 2015 – May 2016

PUBLICATIONS

10. **Baker LJ**, Chávez-Dorado JE, Aggarwal A, Garrey I, & DiBenedetto MH. “The importance of particle shape on sunlight exposure in a wind-driven, free-surface flow.” *Under review*.
9. Chávez-Dorado JE, **Baker LJ**, Riley J, & DiBenedetto MH. “Turbulent diffusivity and effective rise velocity of buoyant particles in a free-surface boundary layer.” *Under review*.
8. Sanness Salmon H, **Baker LJ**, Kozarek J, & Coletti F (2023) “Effect of Shape and Size on the Transport of Floating Particles on the Free-surface of a Meandering Stream.” *Water Resources Research*, 59, e2023WR035716.
7. **Baker LJ** & DiBenedetto MH (2023) “Large-scale particle shadow tracking and orientation measurement with collimated light.” *Experiments in Fluids*, 64, 52.
6. **Baker LJ** & Coletti F (2022) “Experimental investigation of inertial fibres and disks in a turbulent boundary layer.” *Journal of Fluid Mechanics*, 943, A27.
5. **Baker LJ**, Qiao Y, Ghaemi S, & Coletti F (2021) “Method to minimize polymer degradation in drag-reduced non-Newtonian turbulent boundary layers.” *Measurement Science and Technology*, 32, 085303.
4. **Baker LJ** & Coletti F (2021) “Particle–fluid–wall interaction of inertial spherical particles in a turbulent boundary layer.” *Journal of Fluid Mechanics*, 908, A39.
3. **Baker LJ** & Coletti F (2019) “Experimental study of negatively-buoyant finite-size particles in a turbulent boundary layer up to dense regimes.” *Journal of Fluid Mechanics*, 866, 598-629.
2. Petersen A, **Baker LJ**, & Coletti F (2019) “Experimental study of inertial particles clustering and settling in homogeneous turbulence.” *Journal of Fluid Mechanics*, 864, 925-970.
1. **Baker LJ**, Frankel A, Mani A, & Coletti F (2017) “Coherent clusters of inertial particles in homogeneous turbulence.” *Journal of Fluid Mechanics*, 833, 364-398.

INVITED TALKS

Mechanical Engineering Graduate Seminar, University of Iowa (Feb 2023)

Workshop on Ocean Microplastic Transport, Banff International Research Station (Feb 2022)

Ocean Engineering Seminar, University of New Hampshire (Feb 2022)

Hydro-Geo Seminar, Univ. of Minnesota & Univ. of Illinois Urbana-Champaign (Feb 2022)

Environmental Fluid Mechanics Seminar, University of Washington (Oct 2021)

MENTORSHIP, OUTREACH, & SERVICE

Peer Review

Mar 2019 – present

Journal referee: *Journal of Fluid Mechanics*, *Experiments in Fluids*, *International Communications on Heat and Mass Transfer*, *European Journal of Mechanics / B Fluids*, *Physical Review Fluids*, *Water Research*, *Microplastics & Nanoplastics*, *Journal of Geophysical Research - Solid Earth*

Proposal reviewer: NSF

Research Mentor, University of Washington

Jun 2021 – Jan 2023

Postdoc Union Steward, University of Washington

May 2022 – Jul 2023

Climate Justice Working Group, University of Washington

Jun 2021 – Jul 2022

Volunteer Income Tax Preparer, United Way of King County, WA

Jan 2023 – Apr 2025

Outreach Volunteer, Seattle Astronomical Society

Feb 2022 – Jan 2023

Peer Mentor, Women of Aeronautics and Astronautics, Univ. of Minnesota

Sep 2017 – Jun 2021

PROFESSIONAL DEVELOPMENT

Leadership Academy and Network for Diversity in the Geosciences, American Geophysical

Union Postdoctoral Research Fellows Program

Jan 2022 – Jan 2023

Space Weather Simulation Summer School, University of Colorado

Jul 2022

Summer Institute on Earth-Surface Dynamics, St. Anthony Falls Lab

Aug 2017

Solar Vehicle Project, University of Minnesota

Sep 2012 – Jan 2015

CONFERENCE PRESENTATIONS

Chávez-Dorado JE, Riley J, **Baker LJ**, & DiBenedetto MH (2026) "Lagrangian dispersion of finite-sized particles under surface waves and turbulence." APS Division of Fluid Dynamics, Orlando, FL.

Chávez-Dorado JE, **Baker LJ**, & DiBenedetto MH (2026) "Buoyant Particle Transport in Free-Surface Boundary Layers." AGU Ocean Sciences Meeting, Glasgow, Scotland.

Chávez-Dorado JE, **Baker LJ**, & DiBenedetto MH (2025) "Vertical transport of buoyant particles under a free-surface boundary layer." APS Division of Fluid Dynamics, Houston, TX.

Chávez-Dorado JE, Maschhoff A, **Baker LJ**, & DiBenedetto MH (2024) "Particle transport under a wind-driven boundary layer." APS Division of Fluid Dynamics, Salt Lake City, UT.

LOU BAKER, PhD

- DiBenedetto MH, Chávez-Dorado JE, & **Baker LJ** (2024) "Dynamics of buoyant, non-spherical particles in a wind-driven surface boundary layer." Ocean Sciences Meeting, New Orleans, LA.
- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2023) "Non-spherical particle orientation dynamics under a wind-driven free surface boundary layer." APS Division of Fluid Dynamics, Washington D.C.
- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2023) "Buoyant, Non-Spherical Particles in Wind-Driven Waves." Waves in Sea Environment Meeting, Princeton, NJ.
- Baker, L**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2022) "Buoyant, Non-Spherical Plastic Particles in Turbulent Wind-Driven Waves." AGU Fall Meeting, Chicago, IL.
- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2022) "Buoyant, non-spherical particles in turbulent wind-driven waves." APS Division of Fluid Dynamics, Indianapolis, IN.
- Baker LJ**, Aggarwal A, Chávez-Dorado JE, Garrey I, & DiBenedetto MH (2022) "Buoyant, non-spherical particles in turbulent wind-driven waves." Atmospheres, Oceans, Earths—Unifying perspectives on geophysical and environmental multiphase flows, KITP, Santa Barbara, CA.
- Axworthy J, Wang S, DiBenedetto MH, **Baker LJ**, & Padilla-Gamiño J (2022) "Microplastics ingestion and adhesion by reef-building corals under different flow rates." International Coral Reef Society, Bremen, Germany.
- Baker LJ** & DiBenedetto MH (2022) "Vertical transport and orientation of buoyant, non-spherical particles in the wind-mixed ocean surface boundary layer." Ocean Sciences Meeting, virtual.
- Baker LJ** & Coletti F (2021) "Orientation and tumbling of inertial rod and disk particles in a turbulent boundary layer." APS Division of Fluid Dynamics, Phoenix, AZ.
- Baker LJ** & Coletti F (2020) "Effects of shape on microplastic particle–fluid–wall interaction and transport in a turbulent boundary layer." AGU Fall Meeting, virtual.
- Baker LJ** & Coletti F (2020) "Particle-fluid-wall interaction of anisotropic inertial particles in a turbulent boundary layer." APS Division of Fluid Dynamics, virtual.
- Sanness Salmon H, **Baker LJ**, Kozarek J, & Coletti F (2020) "Effect of size and shape on the transport of particles over the free surface of a natural stream." APS Division of Fluid Dynamics, virtual.
- Baker LJ** & Coletti F (2019) "Experimental Investigation of the Dynamics of Resuspending Spherical Sediment Particles in a Turbulent Boundary Layer." AGU Fall Meeting, San Francisco, CA.
- Coletti F & **Baker LJ** (2019) "Simultaneous tracking of suspended particles and time-resolved PIV in a turbulent boundary layer." APS Division of Fluid Dynamics, Seattle, WA.
- Petersen A, **Baker LJ**, & Coletti F (2017) "Laboratory Study of Air Turbulence-Particle Coupling." AGU Fall Meeting, New Orleans, LA.
- Baker LJ** & Coletti F (2017) "Experimental study of dense suspension of large particles in a turbulent boundary layer." APS Division of Fluid Dynamics, Denver, CO.
- Petersen A, **Baker LJ**, & Coletti F (2017) "Particle Plumes Falling Through Quiescent and Turbulent Environments." APS Division of Fluid Dynamics, Denver, CO.
- Petersen A, Carter D, **Baker LJ**, & Coletti F (2017) "Experimental Study of Particle-turbulence Interaction in Homogeneous Turbulence." 10th International Symposium on Turbulence and Shear Flow Phenomena, Chicago, IL, USA.
- Coletti F, Toloui M, Fong, KO, Nemes A, & **Baker LJ** (2016) "Volumetric distribution and velocity of inertial particles in a turbulent channel flow." 18th International Symposium on Application of Laser and Imaging Techniques to Fluid Mechanics, Lisbon, Portugal.
- Coletti F, Petersen A, Carter D, & **Baker LJ** (2016) "Measurements of particle settling velocity in homogeneous turbulence with no mean flow." International Conference on Multiphase Flows, Florence, Italy.
- Baker LJ**, Frankel A, Mani A, & Coletti F (2016) "Coherent clusters of inertial particles in homogeneous turbulence." APS Division of Fluid Dynamics, Portland, OR.
- Petersen A, Carter D, **Baker LJ**, & Coletti F (2015) "Settling of inertial particles through quiescent, weakly turbulent and strongly turbulent air." APS Division of Fluid Dynamics, Boston, MA.